Simulation Tutorials
Lecture 1

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1.1. Purpose and Important Points of This Tutorial

Purpose:

- The purpose of this tutorial is to develop practical knowledge and skills in computer simulations, which are essential across a wide range of scientific research.
- To this end, the lectures and exercises will cover a broad spectrum of numerical methods, ranging from basic programming for **analysis** and **visualization** using C (C++) and Python, to practical simulation techniques such as the Monte Carlo method and molecular dynamics, with an emphasis on their applications in scientific research.

Important Points:

- A self-study assignment will be provided after each tutorial session.
- The main computations in this tutorial will be demonstrated using C (C++), while visualization will be performed using Python, with detailed explanations.
- However, for the self-study and report assignments, you are free to use other programming languages or adjust the workflow (e.g., implementing everything in Python).

Sample Programs:

- Sample programs are available in the following GitHub repository. Please download them as needed.
  Link

1.2. Syllabus

The following topics are planned to be covered in this tutorial (subject to change depending on the progress).

1. Introduction
   - Basics of C (C++) (mainly for numerical calculations)
   - Basics of Python (for data analysis and visualization)
   - Concepts of numerical calculations
   - Loss of significance
   - Non-dimensionalization in scientific calculations
2. Numerical solutions of ordinary differential equations: Examples with harmonic oscillators and damped oscillations
   - Numerical integration of differential equations
   - Stability and conservation laws of orbits
3. Brownian motion of a single particle
   - Langevin equation (stochastic differential equation)
   - Generation of normal random numbers
   - Euler–Maruyama method
   - Time averaging and ensemble averaging
4. Brownian motion of multi-particle systems
   - Calculation methods for interaction forces
   - Non-equilibrium system simulations: example of phase separation
5. Molecular dynamics simulation of multi-particle systems
   - Position Verlet method and velocity Verlet method
   - Conservation laws in multi-particle systems
6. Monte Carlo method
   - Review of statistical mechanics
   - Markov chain Monte Carlo method
   - Metropolis algorithm

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2.1. Setting up the C(C++) Environment

- In this tutorial, the primary numerical calculations will be done using C(C++), and plotting will be done using Python.
- Setting up the C(C++) environment:
  - **Windows:** Installing Ubuntu Dr. Yoshii's manual (in Japaneses): [**Link**] [2]
  - **mac:** Installing Homebrew [**Homebrew Official Link**] [2]

List 1: (On terminal) (For mac) Installing emacs (terminal editor) using Homebrew.

```
brew install emacs
```

Common Unix Commands

- cd (directory name): Change directory
- mkdir (directory name): Create a directory
- rm (directory name): Remove a directory
- touch (file name): Create a file
- ls: List files and directories

2.2. Compiling C(C++) Programs

This section explains how to compile and run the following C(C++) program (pi.cpp) on your computer.

- **C Compiler:**

Various Compilers

- gcc: GNU C compiler collection.
- g++: GNU C++ compiler collection.
- clang: An alternative to gcc — the default compiler on macOS.
- clang++: An alternative to g++ — the default compiler on macOS.
- icc: Intel C/C++ compiler — expensive but offers faster computation speed.

Compilation Options

- -O3: One of the optimization options (i.e., -O -O0 -O1 -O2 -O3 -Os -Ofast -Og). -O3 is most commonly used.
- -o: Specifies the name of the output file.

- Sample program “pi.cpp”

2.2. Compiling C(C++) Programs (2)

List 2: Sample program in C “pi.cpp”

```
#include <stdio.h> // for printf, etc
#include <stdlib.h> // for rand(), etc
#include <math.h> // for sin(),cos(), etc
#include <iostream>// for cout, etc
#include <iomanip>// for setprecision()
#include <fstream> // for ifstream/ofstream
#include <time.h>// for time(NULL), etc

int main(void){
  int i, count = 0, max = 1e+5;
  double x,y,z,pi;
  char fname[128];
  std::ofstream file;
  srand(time(NULL));  // "time(NULL)" as a seed of ramdom number
  sprintf(fname,"coord%d.dat",max); // Define the file name for fname[128]
  file.open(fname); // "file" with the name of fname[128]
  for(i=0;i<max;i++){
    x = (double)rand()/RAND_MAX;
    y = (double)rand()/RAND_MAX;
    z = x*x + y*y;
    if(z<1){
      count++;
```

2.2. Compiling C(C++) Programs (3)

```
      file << x <<" "<<y <<std::endl;
    }
  }
    file.close();
    pi = (double)count / max * 4.;
    printf("%.20f\n",pi); //by C, %.20f -- Displaying with 20 decimal precision
    std::cout<< std::setprecision(21) <<  pi  << std::endl; // by C++

    return 0;
}
```

- Download the program from the designated GitHub repository [**Link**] [1].
- How to download from GitHub [**Link**] [3].
- After downloading, place ‘pi.cpp‘ in an appropriate directory.
- In the directory where ‘pi.cpp‘ is located, execute the following commands.
- Compilation method:

2.2. Compiling C(C++) Programs (4)

List 3: （On terminal） Example of code compilation:（**execute each**）. Use the -o option to specify the name of the output executable file. a.out is obtained in case of no -o option.

```
g++ -O3 pi.cpp
g++ -O3 pi.cpp -o pi.out
g++ -O3 -o pi.out  pi.cpp
clang++ -O3 -o pi.out  pi.cpp
```

- Execution method:

List 4: (Linux/Mac terminal) Running the program (pi.out).

```

./pi.out
```

- Execution result:

List 5: (On terminal) Example of program execution result. Note that the values may vary each time as the random seed is set to the current time.

```

3.14536000000000015575
3.14536000000000015575
```

2.2. Compiling C(C++) Programs (5)

Useful Editor (Optional): Visual Studio Code

Setup Manual (in Japanses, Created by Mr. Ikeda, R Lab, Nagoya):

- Win: [**Link**]
- Mac: [**Link**]

2.3. Setting up the Python Environment

- In this course, Python programs will be executed using Jupyter Notebook.
- Jupyter Notebook is included in Anaconda Navigator ([**Official Link**], see Figure **1**).
- Refer to the following for instructions on installing Anaconda Navigator.
  - **Windows/mac:** Installation of Anaconda Navigator
    [**Windows manual**] [**Mac manual**]
- The installation method for Jupyter Notebook is shown in Figure **2**.

2.3. Setting up the Python Environment (2)

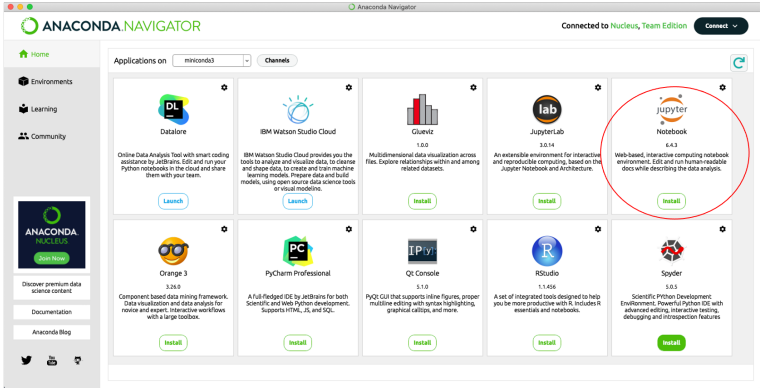
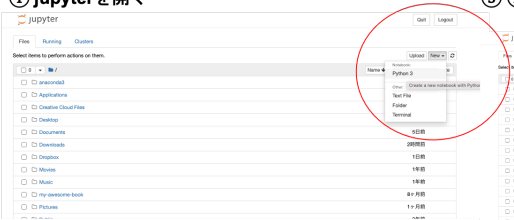


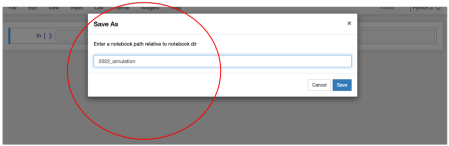
Fig 1: Install and run the Jupyter note book (circled in red) found in the Anaconda Navigator. Now you can use python.

2.3. Setting up the Python Environment (3)

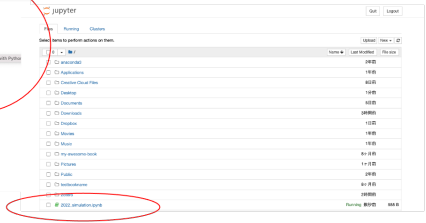
① jupyterを開く



② Newのタブを開きpython3を選び，新規Notebookに名前をつける．



③ ②で作ったnotebookを開く



④ グレーの箱にプログラムを書き込み，shift+enterで実行

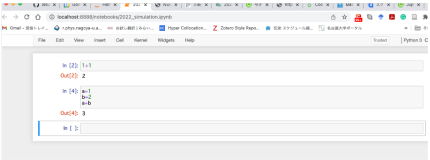


Fig 2: From the start of the Jupyter notebook to the execution of python..

2. 4. Running Python (matplotlib)

- Run the following program (plotting using matplotlib.pyplot) on Jupyter Notebook (see the execution result in Figure **3**).
- (Reference) matplotlib manual [**Link**]

Supplement on Plotting with Python [4]

- matplotlib: A graphics package for Python.
- matplotlib.pyplot: Provides various graph tools such as axes, figure, plot, scatter, etc.

List 6: Importing and using matplotlib.pyplot example

```
import matplotlib.pyplot as plt
fig = plt.figure(figsize=(7,7))
```

- NumPy: Provides fast computation. Many packages refer to NumPy. Offers various linear algebra operations, statistical operations, special functions, etc.

List 7: Importing NumPy

```
import numpy as np
```

2. 4. Running Python (matplotlib) (2)

List 8: Python Sample Program “matplot.py”

```
import matplotlib.pyplot as plt
%matplotlib inline
# Increase the resolution of the graph
%config InlineBackend.figure_format = 'retina'
price = [100, 250, 380, 500, 700]
number = [1, 2, 3, 4, 5]

# Plot the graph
plt.plot(price, number)

# Graph title
plt.title("price / number")

# x-axis label
plt.xlabel("price")

# y-axis label
plt.ylabel("number")

# Display the graph
plt.show()
```

2. 4. Running Python (matplotlib) (3)

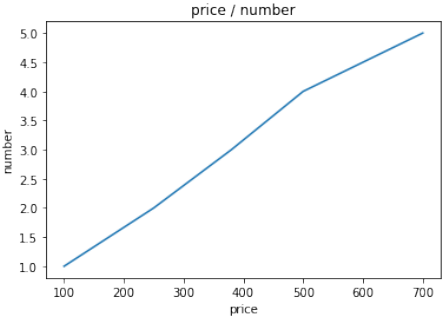


Fig 3: Execution result of matplot.py.

- Next, create a scatter plot of “coord100000.dat” generated by running the program in List 2, as shown below.
- In the sample program “coord.py,” some advanced formatting is applied, such as adjusting the size of the figure, aspect ratio, axis thickness, font size, and font (TeX). Execution result (Figure **4**):

2. 4. Running Python (matplotlib) (4)

List 9: Python Sample Program “coord.py”

```
import matplotlib
import matplotlib.pyplot as plt
%matplotlib inline
# Increase the resolution of the graph
%config InlineBackend.figure_format = 'retina'
import numpy as np
# TeX font
#plt.rcParams["text.usetex"] =True

# Change the overall size and aspect ratio of the figure
fig = plt.figure(figsize=(7,7))

# When placing multiple graphs, change this
ax = fig.add_subplot(111)

# Change the file path as needed
x, y  = np.loadtxt("./Lecture1/coord100000.dat", comments='#', unpack=True)
plt.plot(x, y, "o",markersize=0.5,color="b",label=r"$x^2+y^2\leq1$")

# Graph formatting
plt.tick_params(which='major',width = 1, length = 10)
plt.tick_params(which='minor',width = 1, length = 5)
```

2. 4. Running Python (matplotlib) (5)

```
ax.spines['top'].set_linewidth(3)
ax.spines['bottom'].set_linewidth(3)
ax.spines['left'].set_linewidth(3)
ax.spines['right'].set_linewidth(3)
plt.xlabel(r"$x$",color='k', size=30)
plt.ylabel(r"$y$",color='k', size=30)
plt.xticks(color='k', size=25)
plt.yticks(color='k', size=25)
# Adjust the presence, position, and size of the graph legend
plt.legend(ncol=1, loc=1, borderaxespad=0, fontsize=25,frameon=True)
# Set graph margins
plt.subplots_adjust(wspace=0.0, hspace=0.25)
# Set the aspect ratio of each graph to 1:1
ax.set_aspect('equal', adjustable='box')
# Change the file path as needed.
plt.savefig('./Lecture1/coord.png')
plt.savefig('./Lecture1/coord.pdf')
```

2. 4. Running Python (matplotlib) (6)

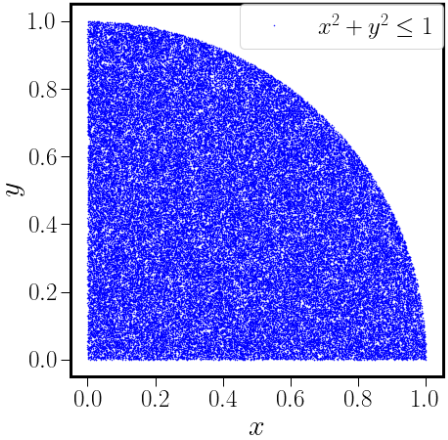


Fig 4: Execution result of coord.py.

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3. First Assignment

Convergence of π (Monte Carlo Simulation)

(1) Based on the sample program in List 2, examine the convergence of the calculated value of π as the number of random samples n changes. Create a graph with n on the horizontal axis and the calculated value of $\pi(n)$ on the vertical axis. Also, plot the error $\delta(n) = |\pi(n) - \pi|$ relative to the theoretical value of π as a function of n, and evaluate the results.

(2) (Advanced Problem) In large-scale numerical simulations, the standard pseudo-random number generator rand() may occasionally cause issues due to its short period. On the other hand, the Mersenne Twister, developed by Makoto Matsumoto, is known for its high-quality pseudo-random number generation [**Reference Link**] [6]. Modify the sample program in List 2 to use the Mersenne Twister instead [**Reference Link**] [7]. The Mersenne Twister header file has been placed in the designated GitHub repository [**Link**] [1], so feel free to use it.

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References and Websites

[1] Takeshi Kawasaki.
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https://tetsufuku-blog.com/github-download/, April 2020.

[4] Hiroshi Hashimoto and Koji Makino.
Python コンピュータシミュレーション入門.
オーム社, 2021.

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Mersenne Twister: A random number generator (since 1997/10).
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[7] Takahiro Ohmi.
C 言語による乱数生成.
https://omitakahiro.github.io/random/random_variables_generation.html.